

Lecture 8

Vague and Uncertain Knowledge

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Today's lecture is an introduction to vagueness and uncertainty and how/when we use fuzzy sets and probability in AI.

Overview

- 1 Uncertainty versus Vagueness
- 2 Fuzzy Logics
- 3 Basic of Probability

Part 0: Vagueness versus Uncertainty

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Uncertainty and Vagueness are closely related terms, and often confused. Let us take a closer look at the difference between the two.



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Is the glass half-empty or half-full? We can all agree that it has half water and half air (even though there might possibly also be some uncertainty about this).

Note that we have in principle (almost) full observability, so we could check whether this glass is really half full or not by some kind of sensors.

A photograph of a clear glass filled with water, sitting on a wooden surface. The glass is about half full. A speech bubble points from the text 'Seems more to me' to the water level in the glass. The background is a solid orange color.

Seems more
to me

But, is it really half? Maybe it is not exactly half. Maybe only roughly half. This is a vague notion of how full the glass is. In fuzzy terminology one would say that the glass is half full to a very high degree.

This has nothing to do with observability.



Now the room gets dark. How much water is in the glass? We simply do not know, there is partial or no observability



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Now the room gets dark. How much water is in the glass?

Uncertainty versus Vagueness



Uncertainty



Vagueness

Uncertainty and vagueness are different notions with different goals.

There are also ways to express vagueness with probabilities, but for the sake of simplicity, we look at the most prominent formalisms separately, one focussing on modelling vague information (fuzzy sets and logics), one on uncertainty (probabilities).

Part 1.1: Fuzzy Sets and Vagueness

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We will start with formalisms to model vague knowledge and reason with it.

Classical naive sets

Model theory is often based on Set Theory

- the set $A = \{a, b, c\}$
- the set of tall people : $\{x \text{ is a person} \mid \text{length}(x) \geq 1\text{m}90\}$



- the set of all even numbers: $\{x \mid x \bmod 2 = 0\}$

Remember what we had seen when modeling with Propositional Logics. A statement (proposition) could only be true or false, e.g. a tall person would be 1m90 or taller, etc. This means someone with 1m89 would not be tall, and as not-tall as someone with 1m52.

Vagueness

Fuzzy set theory allows an object to be *to some degree* an element of a (fuzzy) set.

- the set of tall people : $\{x \text{ is a person} \mid x \text{ is tall}\}$
 - $\{1.90, 1.20, 1.85, 1.70, 1.60, 2.00, \dots\}$

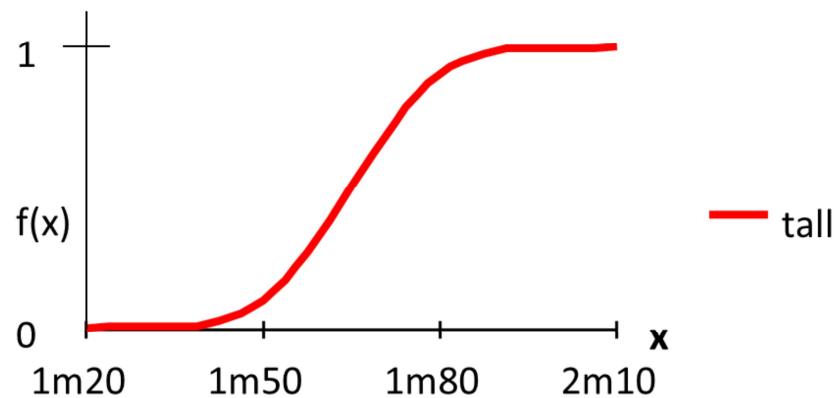


We need to define
what this means!!

Presently, **fuzzy set theory** is the dominant approach to dealing with vague terms.

In a way we would like to model that someone who is 170cm tall is to a higher degree tall than someone who is 120cm tall.

Example of Fuzzy Membership Function



Graph of the membership function of the fuzzy set of tall people.

This notion of a fuzzy membership is normalised to values between 0 and 1.

This is very common (and sometimes a bit confusing). Normalised results can be more easily compared within the formalisms.

However, this does not mean that values can be compared between different formalisms.

0.5 tall and 0.5 certain that someone is tall is something completely different.

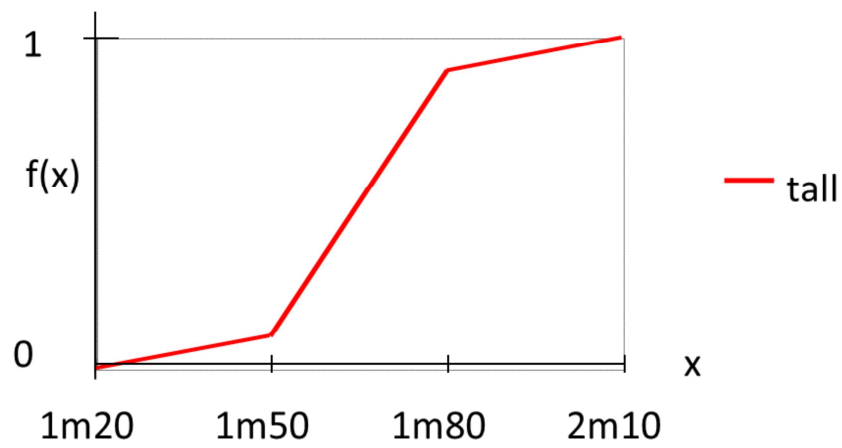
Fuzzy Sets

Universe U , object $x \in U$

A (fuzzy) membership function for a fuzzy set A is defined to be a function f_A from U to $[0,1]$.

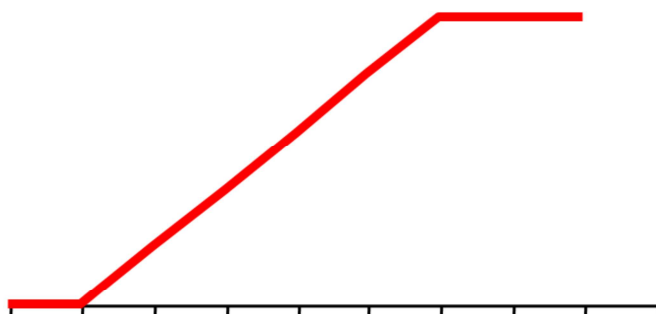
- $f_A(x) = y$: x is a member of A to degree y .
 - Alternative reading: A holds for x with certainty y (vagueness via uncertainty!)
- $f_A(x) = 1$: x is a member of A (*certainly*).
- $f_A(x) = 0$: x *not* a member of A (*certainly*).

Piecewise Linear Approximations

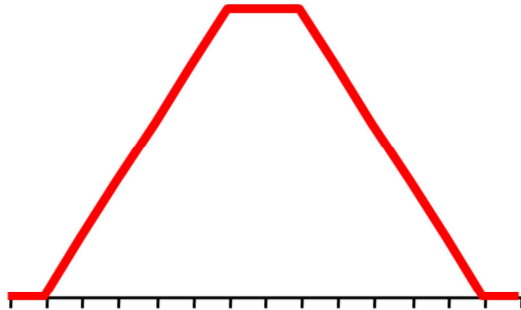


Often, piecewise linear approximations are used.

Increasing Notions



Approximating Notions



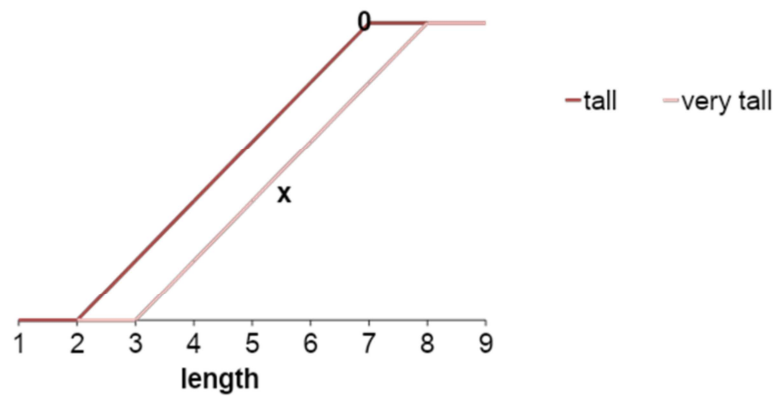
Shape of graph of a vague *approximating notion*, like “medium-sized”, “near Amsterdam”, “about average”, “nearly half-full glass” etc.

How would you model *non-average result* of an exam?

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But there are also other types of shapes for fuzzy functions. Think of notions of “nearly half-full”, which are 0 when empty, 1 when half full and 0 again when fully filled.

Modifiers (Hedges)



“very” can be viewed as a simple modifier of a fuzzy set: move the graph of the membership function to the right

Operations on Fuzzy Sets

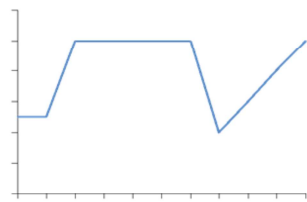
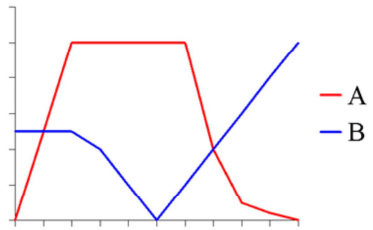
- Complement $f_{\neg A}(x) = 1 - f_A(x)$
- Union $f_{A \cup B}(x) = \max(f_A(x), f_B(x))$
 $f_{A \cap B}(x) = \min(f_A(x), f_B(x))$
- Intersection $A \subseteq B \Leftrightarrow \forall x (f_A(x) \leq f_B(x))$

Reduces to the familiar notions if the sets are **crisp**.

- Alternative choices are possible, and also used.

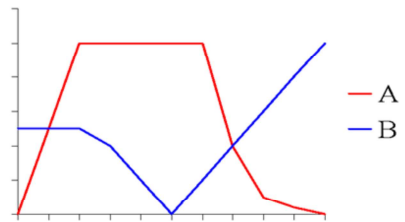
For example: product instead of min.

Union

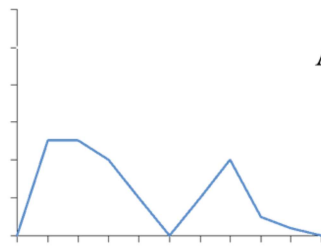


$A \cup B$

Intersection



$A \cap B$



$$f_{A \cap B}(x) = \min(f_A(x), f_B(x))$$

Part 1.2: Fuzzy Logic

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On the basis of Fuzzy Sets, we can now also define a logic.

Fuzzy Logic

Syntax is simply Propositional Logic syntax

The *multivalued logic* corresponding to the defined set-operations. Let $I(.)$ be an interpretation function such that

- $I(A)$ is a fuzzy value $[0,1]$ for all A
- $I(\neg A) = 1 - I(A)$
- $I(A \vee B) = \max(I(A), I(B))$
- $I(A \wedge B) = \min(I(A), I(B))$
- $I(A \rightarrow B) = \min(1, 1 - I(A) + I(B))$

More general: *reasoning with fuzzy sets*.

- This includes several aspects not present in classical set theory: modifiers (a.k.a. hedges), fuzzy composition, etc.

Fuzzy Logic

The meaning of a sentence is determined by an interpretation function that assigns a fuzzy set value to propositions and extends:

- $I(\neg A) = 1 - I(A)$
- $I(A \vee B) = \max(I(A), I(B))$
- $I(A \wedge B) = \min(I(A), I(B))$
- $I(A \rightarrow B) = \min(1, 1 - I(A) + I(B))$

Exercise: Suppose

$$I(P)=0.4,$$

$$I(Q)=0.3$$

what is the meaning, i.e. the fuzzy value of $I(P \rightarrow (\neg Q \rightarrow P))$

Part 1.3: Fuzzy Relations

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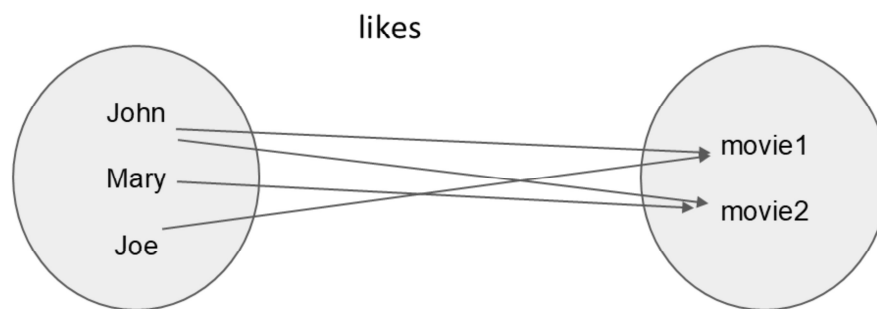
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Why do we care about uncertainty?

Standard Relations

Relation between pairs from two sets

$A = \{\text{John, Mary, Joe}\}$ $B = \{\text{movie1, movie2}\}$



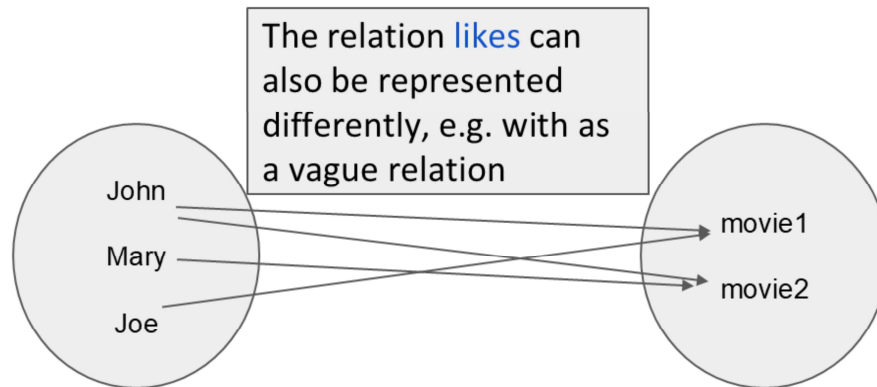
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Remember that we had defined relations (a month ago), as subsets of the Cartesian Product. So, relations are defined depending on the definition of Sets. Now that we have a fuzzy notion of sets, it also makes sense to extend the notion of relations to fuzzy relations.

Standard Relations

Relation between pairs from two sets

$A = \{\text{John, Mary, Joe}\}$ $B = \{\text{movie1, movie2}\}$



Standard Relations

Relation between pairs from two sets

$A = \{\text{John, Mary, Joe}\}$ $B = \{\text{movie1, movie2}\}$

$\text{likes} = \{(\text{John, movie1}), (\text{John, movie2}), (\text{Mary, movie2}), (\text{Joe, movie1})\}$

likes =

	movie1	movie2
John	1	1
Mary	0	1
Joe	1	0

Fuzzy Relations

Fuzzy sets can denote **fuzzy properties** of 1 object

- tall, low, wet, feverish, etc.

Fuzzy sets can also denote **fuzzy relations** between multiple objects.

- approximately equal, close to, much larger than, likes, etc.

Fuzzy Relations

Relation between pairs from two sets

$A = \{\text{John, Mary, Joe}\}$ $B = \{\text{movie1, movie2}\}$

$\text{likes} = \{(\text{John, movie1}), (\text{John, movie2}), (\text{Mary, movie2}), (\text{Joe, movie1})\}$

likes =

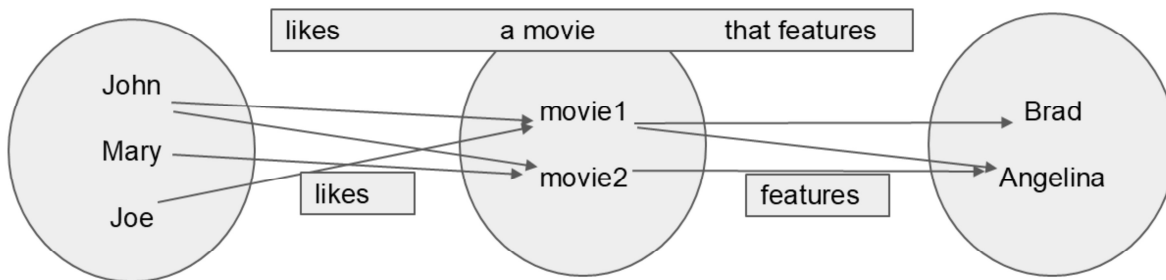
	movie1	movie2
John	0.7	0.8
Mary	0	1
Joe	0.3	0

Composition of Standard Relations

$R \circ S$ stands for the **composition** of R and S (sometimes also $R ; S$)

$$\left. \begin{array}{l} R \subseteq X \times Y \\ S \subseteq Y \times Z \end{array} \right\} \Longrightarrow R \circ S \subseteq X \times Z$$

Classical: $\langle x, z \rangle \in R \circ S \Leftrightarrow \exists y (\langle x, y \rangle \in R \wedge \langle y, z \rangle \in S)$



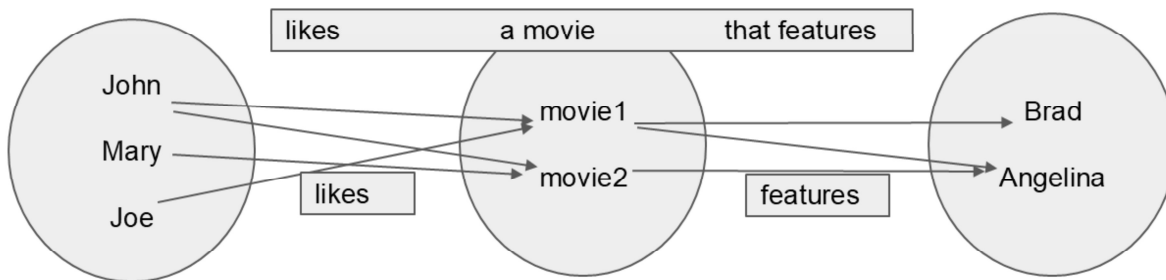
Fuzzy Composition

$R \circ S$ stands for the **composition** of R and S (sometimes also $R ; S$)

$$\left. \begin{array}{l} R \subseteq X \times Y \\ S \subseteq Y \times Z \end{array} \right\} \Longrightarrow R \circ S \subseteq X \times Z$$

Fuzzy composition

$$f_{R \circ S}(\langle x, z \rangle) = \max_{y \in Y} \min(f_R(\langle x, y \rangle), f_S(\langle y, z \rangle))$$



Fuzzy Composition (example)

likes

	movie1	movie2
John	0.7	0.8
Mary	0	1
Joe	0.3	0

features

	Brad	Angelina
movie1	0.3	0.7
movie2	0	0.9

Fuzzy Composition (example)

likes

	movie1	movie2
John	0.7	0.8
Mary	0	1
Joe	0.3	0

features

	Brad	Angelina
movie1	0.3	0.7
movie2	0	0.9

LikesMovieThatFeatures $\approx$ LikesActor

	Brad	Angelina
John	?	?
Mary	?	?
Joe	?	?

Fuzzy Composition (example)

likes

	movie1	movie2
John	0.7	0.8
Mary	0	1
Joe	0.3	0

features

	Brad	Angelina
movie1	0.3	0.7
movie2	0	0.9

Fuzzy composition

$$f_{\text{RoS}}(\langle x, z \rangle) = \max_{y \in Y} \min(f_R(\langle x, y \rangle), f_S(\langle y, z \rangle))$$

	Brad	Angelina
John	?	?
Mary	?	?
Joe	?	?

?

Fuzzy Composition (example)

likes

	movie1	movie2
John	0.7	0.8
Mary	0	1
Joe	0.3	0

features

	Brad	Angelina
movie1	0.3	0.7
movie2	0	0.9

Fuzzy composition

$$f_{\text{RoS}}(\langle x, z \rangle) = \max_{y \in Y} \min(f_R(\langle x, y \rangle), f_S(\langle y, z \rangle))$$

	Brad	Angelina
John	?	?
Mary	?	?
Joe	?	?

max (???,???)

Fuzzy Composition (example)

likes

	movie1	movie2
John	0.7	0.8
Mary	0	1
Joe	0.3	0

features

	Brad	Angelina
movie1	0.3	0.7
movie2	0	0.9

Fuzzy composition

$$f_{\text{RoS}}(\langle x, z \rangle) = \max_{y \in Y} \min(f_R(\langle x, y \rangle), f_S(\langle y, z \rangle))$$

	Brad	Angelina
John	?	?
Mary	?	?
Joe	?	?

max (min (0.7, 0.3), ???)

Fuzzy Composition (example)

likes

	movie1	movie2
John	0.7	0.8
Mary	0	1
Joe	0.3	0

features

	Brad	Angelina
movie1	0.3	0.7
movie2	0	0.9

Fuzzy composition

$$f_{\text{RoS}}(\langle x, z \rangle) = \max_{y \in Y} \min(f_R(\langle x, y \rangle), f_S(\langle y, z \rangle))$$

	Brad	Angelina
John	?	?
Mary	?	?
Joe	?	?

$$\max(\min(0.7, 0.3), \min(0.8, 0))$$

Fuzzy Composition (example)

likes

	movie1	movie2
John	0.7	0.8
Mary	0	1
Joe	0.3	0

features

	Brad	Angelina
movie1	0.3	0.7
movie2	0	0.9

Fuzzy composition

$$f_{\text{RoS}}(\langle x, z \rangle) = \max_{y \in Y} \min(f_R(\langle x, y \rangle), f_S(\langle y, z \rangle))$$

	Brad	Angelina
John	0.3	?
Mary	?	?
Joe	?	?

$$\max(\min(0.7, 0.3), \min(0.8, 0)) = 0.3$$

Fuzzy Composition (example)

likes

	movie1	movie2
John	0.7	0.8
Mary	0	1
Joe	0.3	0

features

	Brad	Angelina
movie1	0.3	0.7
movie2	0	0.9

Fuzzy composition

$$f_{\text{RoS}}(\langle x, z \rangle) = \max_{y \in Y} \min(f_R(\langle x, y \rangle), f_S(\langle y, z \rangle))$$

	Brad	Angelina
John	0.3	?
Mary	?	?
Joe	?	?

Fuzzy Composition (example)

likes

	movie1	movie2
John	0.7	0.8
Mary	0	1
Joe	0.3	0

features

	Brad	Angelina
movie1	0.3	0.7
movie2	0	0.9

Fuzzy composition

$$f_{\text{RoS}}(\langle x, z \rangle) = \max_{y \in Y} \min(f_R(\langle x, y \rangle), f_S(\langle y, z \rangle))$$

	Brad	Angelina
John	0.3	0.8
Mary	0	0.9
Joe	0.3	0.3

More complex Example of Composition

$$f_{\text{ROS}}(\langle x, z \rangle) = \max_{y \in Y} \min(f_R(\langle x, y \rangle), f_S(\langle y, z \rangle))$$

x is much larger than

z	1	2	3	4	5
x \ y					
5					
1	0	0	0	0	0
0					
2	.1	0	0	0	0
.1					
3	.4	.1	0	0	0
.3					
4	.8	.3	.1	0	0
.7					
5	1	.6	.2	.1	0
1					

y is close to

z	1	2	3	4
y \ z				
1	1	.7	.3	.1
2	.7	1	.7	.3
3	.3	.7	1	.7
4	.1	.3	.7	1
5	0	.1	.3	.7

x is much larger than some number close to z

z	1	2	3	4	5
x \ z					
1	0	0	0	0	0

$\min(.8, .7) = .7$
 $\min(.3, 1) = .3$
 $\min(.1, .7) = .1$
 $\min(0, .3) = 0$
 $\min(0, .1) = 0$
 $\max(.7, .3, .1, 0, 0) = .7$

Evaluation of Fuzzy Set Theory

- + Flexible formalism.
- + Coincides with classical set theory for crisp sets.
- + Several successful applications of fuzzy control.
- Requires many arbitrary choices (what are the right operations? Where do the numbers come from?)
- Tends to blur the difference between probabilistic uncertainty, ambiguity, vagueness, etc.

Fuzzy set theory dominant approach to vagueness

- in large part because there is no real alternative.

Part 2: Uncertainty

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Why do we care about uncertainty?

Rational Agents & Uncertainty



A digital airport departure board showing flight information. The board is titled 'DEPARTURES' and has columns for 'Flight', 'To', 'Time', 'Gate', and 'Status'. The time shown is 11:12. The board lists several flights, many of which are marked as 'CANCELLED'.

Flight	To	Time	Gate	Status
ND413	CHAMBERSBURG	1000	1000	CANCELLED
ND219	WINDSOR	1000	1000	New Departs At 10:40
ND227	SLIDERS	1000	1000	CANCELLED
ND228	FLUGELAND	1000	1000	CANCELLED
ND229	CHAMBERSBURG	1000	1000	New Departs At 11:30
ND230	TRAPID	1100	1000	Go to Departure Lounge
ND231	WINDSOR	1100	1000	CANCELLED
ND232	SLIDERS	1100	1000	CANCELLED
ND233	FLUGELAND	1100	1000	New Departs At 12:30
ND234	CHAMBERSBURG	1100	1000	CANCELLED
ND235	TRAPID	1100	1000	CANCELLED
ND236	WINDSOR	1100	1000	CANCELLED
ND237	SLIDERS	1100	1000	CANCELLED
ND238	FLUGELAND	1100	1000	CANCELLED
ND239	CHAMBERSBURG	1100	1000	CANCELLED
ND240	TRAPID	1100	1000	CANCELLED
ND241	WINDSOR	1100	1000	CANCELLED
ND242	SLIDERS	1100	1000	CANCELLED
ND243	FLUGELAND	1100	1000	CANCELLED
ND244	CHAMBERSBURG	1100	1000	CANCELLED
ND245	TRAPID	1100	1000	CANCELLED
ND246	WINDSOR	1100	1000	CANCELLED
ND247	SLIDERS	1100	1000	CANCELLED
ND248	FLUGELAND	1100	1000	CANCELLED
ND249	CHAMBERSBURG	1100	1000	CANCELLED
ND250	TRAPID	1100	1000	CANCELLED
ND251	WINDSOR	1100	1000	CANCELLED
ND252	SLIDERS	1100	1000	CANCELLED
ND253	FLUGELAND	1100	1000	CANCELLED
ND254	CHAMBERSBURG	1100	1000	CANCELLED
ND255	TRAPID	1100	1000	CANCELLED
ND256	WINDSOR	1100	1000	CANCELLED
ND257	SLIDERS	1100	1000	CANCELLED
ND258	FLUGELAND	1100	1000	CANCELLED
ND259	CHAMBERSBURG	1100	1000	CANCELLED
ND260	TRAPID	1100	1000	CANCELLED
ND261	WINDSOR	1100	1000	CANCELLED
ND262	SLIDERS	1100	1000	CANCELLED
ND263	FLUGELAND	1100	1000	CANCELLED
ND264	CHAMBERSBURG	1100	1000	CANCELLED
ND265	TRAPID	1100	1000	CANCELLED
ND266	WINDSOR	1100	1000	CANCELLED
ND267	SLIDERS	1100	1000	CANCELLED
ND268	FLUGELAND	1100	1000	CANCELLED
ND269	CHAMBERSBURG	1100	1000	CANCELLED
ND270	TRAPID	1100	1000	CANCELLED
ND271	WINDSOR	1100	1000	CANCELLED
ND272	SLIDERS	1100	1000	CANCELLED
ND273	FLUGELAND	1100	1000	CANCELLED
ND274	CHAMBERSBURG	1100	1000	CANCELLED
ND275	TRAPID	1100	1000	CANCELLED
ND276	WINDSOR	1100	1000	CANCELLED
ND277	SLIDERS	1100	1000	CANCELLED
ND278	FLUGELAND	1100	1000	CANCELLED
ND279	CHAMBERSBURG	1100	1000	CANCELLED
ND280	TRAPID	1100	1000	CANCELLED
ND281	WINDSOR	1100	1000	CANCELLED
ND282	SLIDERS	1100	1000	CANCELLED
ND283	FLUGELAND	1100	1000	CANCELLED
ND284	CHAMBERSBURG	1100	1000	CANCELLED
ND285	TRAPID	1100	1000	CANCELLED
ND286	WINDSOR	1100	1000	CANCELLED
ND287	SLIDERS	1100	1000	CANCELLED
ND288	FLUGELAND	1100	1000	CANCELLED
ND289	CHAMBERSBURG	1100	1000	CANCELLED
ND290	TRAPID	1100	1000	CANCELLED
ND291	WINDSOR	1100	1000	CANCELLED
ND292	SLIDERS	1100	1000	CANCELLED
ND293	FLUGELAND	1100	1000	CANCELLED
ND294	CHAMBERSBURG	1100	1000	CANCELLED
ND295	TRAPID	1100	1000	CANCELLED
ND296	WINDSOR	1100	1000	CANCELLED
ND297	SLIDERS	1100	1000	CANCELLED
ND298	FLUGELAND	1100	1000	CANCELLED
ND299	CHAMBERSBURG	1100	1000	CANCELLED
ND300	TRAPID	1100	1000	CANCELLED



Example Automatic Taxi Agent

Goal: arrive at airport in time for flight

Environment: traffic, road closure, stop light.

- Not fully observable
- Multi-agent
- Stochastic
- Continuous

How do we address this uncertainty for logical agents?
(of course we have uncertainty in Schnapsen)

Uncertainty is an obstacle for rational agents in dealing with their environment. For example, building an automated taxi agent which takes customers from their homes to the airport requires dealing with uncertainty in many factors; how bad is traffic? Are there any road closures? Will all of the traffic lights be red? These are all part of an environment which is **not fully observable**, meaning the agent can not perceive every relevant factor. It is also a **multi-agent** environment in which other agents are trying to achieve other goals. The environment is **stochastic**, meaning the next state is not observable to the agent, and it is **continuous**.

Sources of Uncertainty

Uncertainty arises from data that is...

1. Unreliable
 - a. Defective measuring instruments
 - b. Biased witnesses
2. Incomplete
 - a. Broken sensor
 - b. Missing laboratory data
 - c. Underspecified information
3. Imprecise
 - a. Approximate measurements
 - b. Limited samples



Uncertainty arises from data which is unreliable, incomplete, or imprecise. In the previous example the data was incomplete, because the environment is not fully observable. Unreliable data can be incorrect, while incomplete data means that the existing data is correct but information is missing. Imprecise measurements may be correct but inadequate for the level of detail required.

Methods for Handling Uncertainty

Let A_{25} mean that we must leave 25 minutes before a flight. Possible interpretations include

Propositional Logic

$A_{25} \ \& \ \neg \text{rain} \ \& \ \neg \text{accident} \ \& \ \neg \text{wind} \ \& \ \dots$
= onTime

Default / Nonmonotonic

Assume my car doesn't have a flat tire.
Assume A_{25} works unless contradicted by evidence.

Logics differ in how they handle contradictions and what assumptions are considered reasonable.

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Logic can handle uncertainty by making statements about what is possible and impossible. Propositional logic would reason about the airport taxi problem by making a series of declarative statements such as "If we leave 25 minutes before the flight and there is no rain, no accidents, no wind, etc., then I will be on time." In default or nonmonotonic logic, the assumption is that everything works positively unless it is contradicted. Different forms of logic are more or less equipped to deal with specific uncertainties.

Early days Methods for Handling Uncertainty

Rules with *fudge factors*:

$A_{25} \rightarrow_{0.3} \text{get there on time.}$

$\text{sprinkler} \rightarrow_{0.99} \text{wetGrass}$

$\text{wetGrass} \rightarrow_{0.7} \text{rain}$

Combining these causes issues!

e.g. sprinkler causes rain?

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Early AI systems, like expert systems, were based on a declarative set of rules such as “If I leave 25 minutes before flight, then in 30% of cases I arrive on time.” These rules are useful in isolation but impossible to combine. For example, if there is wet grass in 99% of cases where the sprinkler is on, how can we differentiate between wet grass due to rain?

Part 2.2: Views on Probability

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Let's review the basic notions of probability, starting with two different philosophical paradigms.

What is the probability of someone having an affair?

1 In 5 Marriages Will Experience An Affair: Here's What Experts Say To Do Next

We asked the experts in family counselling whether it's possible to come back from betrayal. - by Cassandra Green

[f](#) [t](#) [in](#) [p](#) 24 OCT 2023



Frequentist (Objective) Probability

The **frequentist** school of thought says that probability is only a property of repeated experiments and is objective.

newspaper: "one in five people is having an affair"

wrong: the probability that Bob is having an affair is one in five.

right: the probability that a randomly chosen person is having an affair is one in five.

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There are two major schools of thought with respect to probability, and these different schools change the way that a probability is interpreted. Frequentism says that probability is an objective observation that can be generalised to a population. If we observe that 400 students fully participate and pass the course, then this will be true for the next batch of 1000 students as well. It would not be correct to assume that an individual student, Justin, has a 40% chance of fully participating and passing.

Bayesian (Subjective) Probability

The **Bayesian** school of thought says probability is an expression of our uncertainty and beliefs.

“The probability that this die will land on 6 in the next roll is one in six.”

“The probability that Bob is having an affair is one in five.”

To frequentists, the “probability that Bob is having an affair” is wrong because Bob is either having an affair or not. There is no probability involved.

To a Bayesianist, our uncertainty can be expressed as a probability.

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The alternative is called subjective probability, based on the Bayesian paradigm. Bayesianism says that probabilities are an expression of our uncertainty about a specific event, thus not reliant on an observed trend over many experiments. According to Bayesianism, it can not be said that Bob as an individual has a 20% probability of having an affair, because Bob is either having an affair or he is not, and therein lies the uncertainty.

A probabilistic view for passing Intelligent Systems

A probabilistic view on passing the IS course (frequentist/Bayesianist):

WorkGroup	Lecture	Pass	Frequency	Probability
F	F	F	200	0.2
F	F	T	50	0.05
F	T	F	50	0.05
F	T	T	50	0.05
T	F	F	50	0.05
T	F	T	150	0.15
T	T	F	50	0.05
T	T	T	400	0.4

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The frequencies in the table come from observations made in previous years of teaching this course. The frequency is the number of students for whom the conditions on the left were true, for example, 200 students do not attend lectures or workgroups and fail the course. Because the total frequency of students adds up to 1000 students, the probability of a student belonging to each group is the frequency with the decimal point moved three times to the left.

Of course, in reality, I did not do proper bookkeeping, and estimated the probabilities according to my beliefs. So, in truth, this is a rather bayesian view on probabilities, I have to admit.

The good news is that ...

... the mathematics are the same.

Part 2.3: Basics of Probability

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Let's review the basic notions of probability.

Sample Space

The **sample space** of an experiment or trial is a set of all possible outcomes.



$$\Omega = \{\text{heads, tails}\}$$



$$\Omega = \{1, 2, 3, 4, 5, 6\}$$



$$\Omega = \{(1;1), (1;2), (2;1), \dots, (6;6)\}$$

The elements in the set are called sample points, possible worlds, or **atomic events**.

The **sample space** is all the possible outcomes of an experiment, denoted by an omega symbol. For a coin toss, the sample space is only heads or tails because these are the only single outcomes of this experiment. Every single element in this set is called an **atomic event**. In an experiment involving the rolling of two dice, each atomic event is the combination of outcomes of these two dice, that is, every atomic event is a pair.

The Event and Probability Spaces



Event Space $E = \{\{\}, \{1\}, \{2\}, \dots, \{1,2\}, \dots, \{1,2,3,4,5,6\}\}$

Events are...

1. subsets of the sample space (and the things we assign probability to).
2. the atoms in our language.

Unless the sample space is infinite, its powerset $P(\Omega)$ is usually considered the event space.

A **probability space** is a sample space with an assignment $P(w)$ for every atomic event $w \in \Omega$, and event A

- $0 \leq P(w) \leq 1$
- $P(w_1) + P(w_2) + \dots + P(w_n) = 1$ for all $w_i \in \Omega$
- $P(A) = P(w_1) + P(w_2) + \dots + P(w_n)$ for all $w_i \in A$

Note: $\{1,2\}$ stands for either 1 or 2 happened. This is different from $(1;2)$ which we had before.

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Events are the atoms or building blocks of our probability language. Events can be combinations of atomic events or any other subset of the sample space. For a sample space that is finite, meaning there is not an infinite number of possible atomic events, the powerset is considered the event space. Because a powerset always include the empty set, the empty set is also part of the event space.

We can transform a sample space into a probability space by assigning probabilities to each event. The probability value is always between 0 and 1, 0 meaning it never happens and 1 meaning it always happen. The probability for all atomic events sum up to 1, because it is always true that one of the atomic events happens. The probability of a complex event A is defined as the sum of all possible atomic events of the experiment.

Examples of Spaces

Possible outcomes of rolling a die can be represented by sample space
 $\Omega = \{1,2,3,4,5,6\}$

Examples of events

$A_1 = \{2,4,6\}$ “outcome is even”

$A_2 = \{1,2,3,4\}$ “outcome is smaller than 5”

$A_3 = \{6\}$ “outcome is 6”



Example of a probability function

$P(w) = 1/6$ (for each atomic event, i.e. 1, 2, 3, etc.)

Then $P(\{6\}) = 1/6$, $P(\{2,4,6\}) = 1/2$, and $P(\{1,2,3,4\}) = 2/3$

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An event w can be an atomic event itself (e.g. outcome is 6) or a condition (e.g. outcome is even, outcome is smaller than 5). We can assign a probability value to these events $P(w)$ by dividing the number of atomic events in the event by the length of the sample space *if* each atomic event is equally likely. In a dice roll, where each number is equally likely, we can find the chance that the outcome is smaller than 5 by dividing $4/6 = 2/3$ because there are 4 atomic events in which the outcome is smaller than 5 out of 6 possible atomic events.

Random Variables

A **random variable** is a function from atomic events to some **range**, like T/F, real numbers, or discrete values{sunny,rainy}.

e.g. Odd(1) = true, Odd(2) = false, Weather = rainy

P induces a probability distribution $\mathbf{P}(X)$ for all variable values of X

$$P(X = x) = P(w_1) + P(w_2) + \dots \text{ for a random variable } X \text{ and atomic events } w_i \\ \text{where } X(w_i) = x$$

$P(X=x)$ is the probability of an event where the random variable X has value x , and is just the sum of all the atomic events with that property.

Example $P(\text{Odd}=\text{true}) = P(1) + P(3) + P(5) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$

$\mathbf{P}(\text{Odd}) = (\frac{1}{2}, \frac{1}{2})$ as probability distribution for values true and false.

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A random variables is a function that gives an atomic event a particular range. This range can be binary (e.g. true or false), real numbers (e.g. 1, 1.2), or discrete values (e.g. cold, hot). Given the random variable “odd” with range {true, false}, we can say that Odd(1) is true because 1 is an odd number, and Odd(2) is false. Probability P induces a probability distribution for all the random variables. If we want to calculate the probability that the random variable Odd is true, we sum all the individual events to find that there is a $\frac{1}{2}$ probability that a given value is Odd.

$P(X=x)$ is the probability of an event where the random variable X has value x , and is just the sum of all the atomic events with that property.

Language Extension for Boolean Random Variables

A **Boolean proposition** is the event where the proposition is assumed to be true unless negated. If we have Boolean random variables A and B , we can abbreviate:

- Event a = set of atomic events w where
 - $A(w)=\text{True}$
- Event $\neg a$ = set of atomic events w where
 - $A(w)=\text{False}$
- Event $a \wedge b$ = set of atomic events e and f where
 - $A(e)=\text{True}$ and $B(f)=\text{True}$
- Event $a \vee b$ = set of atomic events e and f where
 - $A(e)=\text{True}$ or $B(f)=\text{True}$

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For Boolean random variables denoted with capital letters (A), we assume that the proposition is true unless negated. This allows us to abbreviate the language such that a means that random variable A is true for w , $\neg a$ means A is false, $a \wedge b$ means A and B are true for e and f , and $a \vee b$ means either A or B (or both) are true.

Syntax for Propositions

	Boolean	Discrete	Continuous
Random Variable Range	Odd true/false	Card <diamond,heart,spade,club>	Value 10, 4
Proposition	Odd = true or <i>odd</i>	Card = diamond	Value = 10 Value < 4

For Boolean combination of basic propositions ($\wedge$, $\vee$, $-$, etc.) with related probabilities

$$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$$

e.g. $P(\text{odd} \vee \text{Value} > 4) = P(\text{odd}) + P(\text{Value} < 4) = \frac{1}{2} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6}$

Otherwise written $P(\{1,3,5,6\}) = \frac{4}{6}$

For Boolean, discrete, and continuous variables we denote **propositions**, or true statements, with = to indicate the true value of the variable. For Boolean propositions as shown on the last slide, we can use symbols of propositional logic to combine these statements in formulas.

Intelligent Systems Framework

Efficient, simple, correct, well- understood, generic	Find best decision based on experience.	Weeks 6 & 7
	Find best decision with knowledge & reason.	Weeks 3, 4, 5
	Find best decision in multiple agent setting.	Weeks 1 & 2
	Find best decision with heuristics.	
	Systematically find all possible solutions.	

Progress in the intelligent systems framework.

Summary

Take Home Message

We model uncertain knowledge with **fuzzy sets** and **fuzzy logics**.

We need the notion of **probability** to deal with **uncertainty**.

We can describe probability of **random variables** through sample and event spaces, and organise these probabilities into probability **distributions**.

Thank you for your attention and have a great day.